Andreas Kriegler

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Summary

PhD student in Computer Vision at TU Wien, set to finish in 2027.

I am a PhD student in Computer Vision at the CVL at TU Wien. My supervisor is Prof. Margrit Gelautz. My research is funded by the AIT Austrian Institute of Technology where I am further advised by Csaba Beleznai.

My research lies in the intersection of classical Computer Vision, Deep Learning and Computer Graphics. The goal is to develop novel methods for highly generic object pose estimation. We are specifically interested in the geometry of objects and exploit rendering engines to generate synthetic data.

Before joining the PhD program, I received my M.Sc. and B.Sc. degrees in Mechatronics/Robotics from the UAS Technikum Wien, both with distinction.

Education

TU Wien — *Vienna, Austria*
Dr. techn. (PhD) in Computer Science — *Mar 2021 – present*
- Advisor: Prof. Margrit Gelautz (CVL, Faculty of Informatics, TU Wien)

University of Applied Sciences Technikum Wien — *Vienna, Austria*
M.Sc. (with distinction) in Mechatronics/Robotics — *Sept 2018 – Oct 2020*

University of Applied Sciences Technikum Wien — *Vienna, Austria*
B.Sc. (with distinction) in Mechatronics/Robotics — *Sept 2015 – June 2018*

University of Natural Resources and Life Sciences Vienna — *Vienna, Austria*
B.Sc. in Civil Engineering (discontinued) — *Oct 2014 – June 2015*

Experience

PhD Student — *Vienna, Austria*
AIT Austrian Institute of Technology — *Mar 2021 – present*
Besides the dissertation, I am involved in projects on the automation of large-scale machinery.

Univ. Research Assistant (external) — *Vienna, Austria*
TU WIEN — *Mar 2021 – present*
We are working on a robotic imitation system and corresponding publications for the caring robots project. I am regularly teaching guest lectures and am reviewing external papers as well as internal diploma theses.

Computer Vision Engineer — *Vienna, Austria*
AIT Austrian Institute of Technology — *Nov 2020 – Mar 2021*

Diploma Student — *Vienna, Austria*
AIT Austrian Institute of Technology — *Sept 2018 – Oct 2020*

Application and Control Engineering Intern — *Düsseldorf, Germany*
SMS Group — *Aug 2017 – Jan 2018*

Call Center Agent — *Vienna, Austria*
Customer Care Solutions — *Feb 2016 – Dec 2016*

Paramedic *Vienna, Austria*
Vienna Red Cross *Dec 2013 – Apr 2014*

Volunteer

I Can't Believe It's Not Better Initiative *Worldwide*
I joined the ICBINB initiative after the NeurIPS 2022 workshop. Our aim is to foster a community beyond benchmark climbing, specifically to provide a venue for negative results as well as introspective works. I helped organize the 2023 NeurIPS and 2025 ICLR workshops with them. Together with Prof. Neil Lawrence we publish outstanding works in the Proceedings of Machine Learning Research series. *2022 – present*

Academic work

Teaching (TU Wien, guest lectures)
- Stereo Vision (188.513): 2021S, 2024S
- Video Analysis (188.329): 2021W, 2022W, 2023W
- Visual Analysis of Human Motion (188.468): 2022S, 2023S, 2025S

Reviewing
- Journals: IJCV, IJSR, IJCSS
- Conferences: ICML, SIGGRAPH, DAGM, CVWW, ISPA, ÖAGM

Editor
- NeurIPS/ICLR ICBINB workshop proceedings: PMLR

Summer schools

International Computer Vision Summer School (ICVSS) *Sicily, Italy*
Attended the summer school and passed the examination session (1h) *July 2024*

British Machine Vision Association Computer Vision Summer School *Lincoln, United Kingdom*
Participated in 32 contact hours with exercises at the University of Lincoln *July 2019*

Awards

- Graduated with distinction from the Master's degree (2020) at UAS Technikum Wien
- Graduated with distinction from the Bachelor's degree (2018) at UAS Technikum Wien
- My 2nd Bachelor thesis was nominated as the "Best Bachelor Thesis in Mechatronics in Austria (2018)"

Scholarships

- Received two scholarships from the UAS Technikum Wien for exceptional academic success (17/18, 19/20)
- Received a scholarship from the Karl-Seitz-Stiftung, reserved for students with exceptional academic success and difficult family background (2017)

Certificates

- TOEFL iBT English Language Certificate with grade "C2 (116/120)" (2022)
- Intercultural Skills Certificate from The International Office of the UAS Technikum Wien (2018)
- IAESTE English Language Certificate with grade "1 - excellent" from Berlitz language school (2016)
- Dassault Systemes Mechanical Design (Associate) certification (2016)
- Japanese language courses with certifications of attendance (7.5 ECTS in total, 2015-2017)

Languages

- German (native)
- English (C2)
- Japanese (basic)
- Latin (basic)
- Polish (basic)

References

- Margrit Gelautz, TU Wien, margrit.gelautz@tuwien.ac.at
- Wilfried Wöber, FH Technikum Wien, wilfried.woeber@technikum-wien.at
- Jörg Thomasberger, SMS Group, joerg.thomasberger@sms-group.com

Publications

Towards Symmetry-sensitive Pose Estimation: A Rotation Representation for Symmetric Object Classes Apr 2026
A. Kriegler, C. Beleznai, M. Gelautz
10.1007/s11263-026-02770-x (International Journal of Computer Vision (IJCV))

Review Paper: Body Movement Mirroring and Synchrony in Human-Robot Interaction Oct 2024
D. Stoeva, ***A. Kriegler***, M. Gelautz
10.1145/3682074 (ACM Transactions on Human-Robot Interaction (THRI))

PrimitivePose: Generic Model and Representation for 3D Bounding Box Prediction of Unseen Objects Aug 2023
A. Kriegler, C. Beleznai, M. Gelautz, M. Murschitz, K. Göbel
10.1142/S1793351X23620027 (International Journal of Semantic Computing (IJSC))

Towards Scene Understanding for Autonomous Operations on Airport Aprons Mar 2023
D. Steininger, ***A. Kriegler***, W. Pointner, V. Widhalm, J. Simon, O. Zendel
10.1007/978-3-031-27066-6_11 (16th Asia Conference on Computer Vision (ACCV 2022) - The Fourth International Workshop on Machine Learning and Computing for Visual Semantic Analysis)

PrimitivePose: 3D Bounding Box Prediction of Unseen Objects via Synthetic Geometric Primitives Jan 2023
A. Kriegler, C. Beleznai, M. Murschitz, K. Göbel, M. Gelautz
10.1109/IRC55401.2022.00040 (2022 Sixth IEEE International Conference on Robotic Computing (IRC))

Paradigmatic Revolutions in Computer Vision Dec 2022
A. Kriegler
drive.google.com/file/d/1ItDGcljAQOYo-HKHJLcC4gAgapCIQPns/view (Advances in Neural Information Processing Systems 35 (NeurIPS 2022) - I Can't Believe It's Not Better (ICBINB) Workshop)

Pose-aware object recognition on a mobile platform via learned geometric representations July 2022
C. Beleznai, P. Ausserlechner, ***A. Kriegler***, W. Pointner
10.23919/ASCC56756.2022.9828370 (2022 13th Asian Control Conference (ASCC))

Visual Semantic Context Encoding for Aerial Data Introspection and Domain Prediction Apr 2022
A. Kriegler, D. Steininger, W. Wöber
10.1007/978-3-031-04881-4_34 (Iberian Conference on Pattern Recognition and Image Analysis (IbPRIA) 2022)

Evaluation of Monocular and Stereo Depth Data for Geometry-Assisted Learning of 3D Pose Nov 2021
A. Kriegler, C. Beleznai, M. Gelautz
10.3217/978-3-85125-869-1-01 (Proceedings of the OAGM Workshop 2021)

The Aircraft Context Dataset: Understanding and Optimizing Data Variability in Aerial Domains Nov 2021
D. Steininger, V. Widhalm, J. Simon, ***A. Kiegler***, C. Sulzbacher
10.1109/ICCVW54120.2021.00426 (2021 IEEE/CVF International Conference on Computer Vision Workshops (ICCVW))

Artificial Neural Networks Based Place Categorization Oct 2020
A. Kriegler, W. Wöber, M. Aburaia
10.1007/978-3-030-62784-3_17 (The International Symposium for Production Research (ISPR))

Vision-based Docking of a Mobile Robot Aug 2020
A. Kriegler, W. Wöber
10.3217/978-3-85125-752-6-03 (Joint Austrian Computer Vision and Robotics Workshop (ACVRW))

FH Technikum Wien: Artificial Neural Networks Based State Transition Modeling and Place Categorization Nov 2018
A. Kriegler
mechatronik-plattform.at/wp-content/uploads/2025/04/Tagungsband_2018_FH-Campus02.pdf (Konferenz der Mechatronik Plattform Österreich 2018)

Theses

Visual Semantic Context Encoding for Data Harvesting and Domain Prediction Sept 2020
A. Kriegler, W. Wöber
permalink.obvsg.at/ftw/AC16098872 (Master's Thesis (UAS Technikum Wien))

Artificial Neural Networks based State Transition Modeling and Place Categorization June 2018
A. Kriegler, W. Wöber
www.researchgate.net/publication/352211259_Artificial_Neural_Networks_based_State_Transition_Modeling_and_Place_Categorization (Bachelor's Thesis (2nd, UAS Technikum Wien))

Neuentwicklung der Schlackeauswurferkennung auf der Basis von MATLAB Jan 2018
A. Kriegler, Olaf Nowitzki
www.researchgate.net/publication/352211347_Neuentwicklung_der_Schlackeauswurferkennung_auf_der_Basis_von_MATLAB (Bachelor's Thesis (1st, german, UAS Technikum Wien))